A Project Report

On

"Demand Forecasting Techniques in the Dairy Industry: A Case Study of Amul"

Submitted in Partial Fulfilment of the Requirements for the Degree of

BACHELOR OF BUSINESS ADMINISTRATION

In

Business Management by

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DECLARATION

I hereby declare that the report of the Dissertation Project work titled **"Demand Forecasting Techniques in the Dairy Industry: A Case Study of Amul"** is an authentic record of the project work carried out by me under the supervision of **Dr. Moiz Akhtar** , Assistant Professor, Department of Business Management at Integral University, Lucknow. No part of the project work has been presented elsewhere for any other degree or diploma earlier.

I declare that I have faithfully acknowledged and referred to the works of other researchers, wherever their published works have been cited in this report. I further certify that I have not willfully taken other's work, para, text, data, results, tables, figures, etc. reported in journals, books, magazines, reports, dissertations, theses, etc., or available on the websites, without their permission.

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CERTIFICATE

This is to certify that **Jai Shankar Yadav** (Enrollment No. **2200101061)** has carried out the Dissertation Project entitled **"Demand Forecasting Techniques in the Dairy Industry: A Case Study of Amul"** submitted to the Department of Business Management, Faculty of Commerce & Management, Integral University in partial fulfilment for the award of the Degree of Bachelor of Business Administration from Integral University, Lucknow under my supervision.

It is also certified that

(i) This project work embodies the original work of the candidate and has not been earlier submitted elsewhere for the award of any degree.

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Signature of Supervisor:

Dr. Moiz Akhtar

(Assistant Professor)

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Date:

Place: Lucknow

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INTRODUCTION

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1.1 Overview of the Dairy Industry

The dairy industry is a crucial sector in the global economy, providing essential nutritional products such as milk, cheese, butter, and yogurt. In India, the dairy industry is a significant contributor to the rural economy, employing millions of people directly and indirectly. With an annual production of over 180 million metric tons of milk, India stands as the largest producer and consumer of milk in the world. This booming industry is driven by consumer demand for dairy products, which has been growing steadily over the years. The organized dairy sector in India includes well-established cooperatives like Amul, which has played a key role in shaping the dairy industry and fostering rural development.

1.2 Importance of Demand Forecasting in the Dairy Industry

In an industry like dairy, where products are perishable and demand can fluctuate due to seasonal variations, festivals, and changes in consumer preferences, effective demand forecasting becomes vital for maintaining product availability and minimizing wastage. Demand forecasting techniques help businesses predict future demand for their products, enabling them to optimize production, inventory management, and distribution strategies. For dairy companies like Amul, accurate demand forecasting is essential for meeting the supply-demand gap, ensuring efficient operations, and reducing costs.

Amul, as one of India's leading dairy cooperatives, faces significant challenges in forecasting demand due to the nature of dairy production. These challenges include variability in milk production (which is affected by factors such as climate, feed quality, and cattle health), seasonal demand fluctuations, and changes in consumer buying patterns. Therefore, it is crucial for Amul to implement robust forecasting models to predict demand accurately and ensure product availability across its extensive distribution network.

1.3 Case Study of Amul

Amul is a household name in India, known for its wide range of dairy products such as milk, butter, cheese, ice cream, and milk powders. Founded in 1946, Amul has revolutionized the dairy industry in India and is a shining example of a successful cooperative model. The brand is managed by the Gujarat Cooperative Milk Marketing Federation (GCMMF) and has built a massive supply chain network that reaches millions of customers across urban and rural India. Amul's success lies in its ability to maintain high-quality standards, manage a large producer base, and ensure the smooth functioning of its supply chain, which is heavily dependent on accurate demand forecasting.

Over the years, Amul has adopted various techniques to forecast demand and manage its production and distribution processes efficiently. However, the complexity of the dairy business requires continuous improvement and innovation in forecasting methods to address the evolving challenges. Therefore, this study aims to explore and analyze the demand forecasting techniques employed by Amul, focusing on their effectiveness in ensuring a steady supply of dairy products while minimizing excess inventory and waste.

1.4 Objective of the Study

The primary objective of this study is to evaluate the demand forecasting techniques used in the dairy industry, with a specific focus on Amul. The study will examine the following aspects:

1. **Analysis of Demand Patterns**: Identifying the demand trends and seasonality patterns in the dairy industry, particularly for Amul products.
2. **Forecasting Models**: Reviewing the demand forecasting models implemented by Amul, including time series analysis, machine learning techniques, and other advanced forecasting methods.
3. **Challenges in Demand Forecasting**: Understanding the challenges faced by Amul in forecasting demand, such as fluctuations in milk supply, changing consumer preferences, and external factors like weather conditions and government regulations.
4. **Impact on Operations**: Assessing how accurate demand forecasting impacts Amul's production planning, supply chain management, and inventory control.
5. **Recommendations for Improvement**: Suggesting possible improvements in forecasting methods to enhance efficiency, reduce costs, and improve service levels.

1.5 Significance of the Study

This research holds significant value for the dairy industry, particularly for companies like Amul that play a crucial role in supplying dairy products to millions of customers. By focusing on demand forecasting, this study aims to provide insights into how dairy companies can improve their forecasting accuracy and, consequently, optimize their production and supply chain processes. Furthermore, it will serve as a useful reference for industry practitioners, supply chain managers, and researchers seeking to understand the intricacies of demand forecasting in the dairy sector and apply best practices to enhance operational efficiency.

By exploring Amul's approach to demand forecasting, this case study will contribute to the broader body of knowledge on how large-scale dairy operations can address supply and demand uncertainties and improve business performance through better demand prediction.

1.6 Structure of the Report

This report is structured into several sections. After the introduction, the report will delve into a detailed literature review of existing demand forecasting techniques and their applications in the dairy industry. The methodology section will outline the research methods used to gather data and analyze Amul’s forecasting techniques. Following this, the results and findings will be discussed, followed by an analysis of the effectiveness of Amul’s current forecasting models. Finally, the report will conclude with a summary of the key insights and provide recommendations for improving demand forecasting in the dairy sector.

CHAPTER – 2
REVIEW OF LITERATURE

CHAPTER – 2 REVIEW OF LITERATURE

Demand forecasting is a critical aspect of supply chain management, particularly for industries that deal with perishable goods like dairy. Accurate demand forecasting ensures that production, inventory, and distribution processes are aligned with market requirements, helping to minimize waste and stockouts. This section reviews the existing literature on demand forecasting techniques and models, focusing on the dairy industry and, more specifically, the methods used by major players like Amul.

2.1 Concept of Demand Forecasting

Demand forecasting involves predicting the future demand for a product or service over a specific period. It is essential for decision-making in production planning, inventory management, and distribution. The accuracy of demand forecasting plays a pivotal role in optimizing supply chain operations and reducing operational costs. Inaccurate forecasting can lead to overproduction or underproduction, both of which can have significant financial implications for businesses (Chopra & Meindl, 2019).

In the dairy industry, where products are highly perishable, the importance of demand forecasting becomes even more pronounced. Given the short shelf-life of dairy products, timely and accurate demand forecasting ensures that production levels match consumer demand, minimizing product wastage and ensuring efficient stock management.

2.2 Demand Forecasting Methods

Demand forecasting methods can be broadly classified into two categories: qualitative and quantitative methods.

- **Qualitative Methods**: These methods are based on subjective judgment and expert opinions rather than historical data. They are typically used when there is little historical data available or when there are significant changes in the market environment. Techniques such as Delphi method, market research, and expert judgment are commonly used in the dairy industry for forecasting demand, particularly when entering new markets or launching new products (Makridakis, 1998).
- **Quantitative Methods**: These methods rely on historical data and statistical models to predict future demand. The most commonly used quantitative techniques are time series analysis, causal models, and machine learning algorithms. Time series analysis, for example, is based on the premise that past demand patterns can help predict future demand by identifying trends, seasonality, and cyclical patterns.
    - **Time Series Analysis**: In time series forecasting, past demand data is used to predict future demand by identifying recurring patterns over time, such as weekly, monthly, or seasonal trends. Common models used in time series forecasting

include Moving Averages, Exponential Smoothing, and ARIMA (AutoRegressive Integrated Moving Average) models (Hyndman & Athanasopoulos, 2018). Time series methods are particularly useful in the dairy industry, where demand patterns tend to follow seasonal fluctuations.

- **Causal Models**: These models use external factors or independent variables to predict demand. For example, weather conditions, promotions, or economic factors might influence dairy product demand. Regression analysis is a common technique used in causal models (Wang & Shankar, 2018). For instance, in the dairy industry, temperature and seasonal factors significantly affect milk production, which, in turn, impacts the demand for various dairy products.
- **Machine Learning**: Recently, machine learning and artificial intelligence (AI) have gained traction in demand forecasting. Techniques such as artificial neural networks (ANNs) and support vector machines (SVMs) are now being applied to predict demand based on large volumes of historical data and external variables. Machine learning techniques have the potential to handle non-linear relationships and large datasets, making them highly effective in industries like dairy, where multiple factors influence demand (Bertsimas et al., 2015).

2.3 Demand Forecasting in the Dairy Industry

The dairy industry, like many other sectors, faces unique challenges in demand forecasting. Some of the key challenges include:

- **Seasonal Demand Fluctuations**: Dairy products are subject to significant seasonal variations, with demand rising during festivals and holidays. For example, demand for products like ghee, curd, and ice cream often increases during summer months, while milk consumption may see fluctuations depending on regional festivals and cultural events (Chand, 2019). Accurate forecasting models must account for such variations to prevent overproduction or stockouts.
- **Supply Chain Complexity**: The dairy supply chain involves numerous stakeholders, including farmers, processors, distributors, and retailers. Forecasting demand accurately is crucial to managing the flow of raw materials (milk) and finished products through the supply chain. Moreover, the perishable nature of milk necessitates tight synchronization between production and distribution schedules.
- **Milk Production Variability**: Milk production is influenced by several factors, including cattle health, feed quality, and weather conditions. For instance, adverse weather conditions such as droughts or floods can reduce milk production, leading to supply shortages. Hence, forecasting demand in the dairy industry is not only about predicting consumer demand but also anticipating fluctuations in raw material supply (Nair et al., 2020).

- **Consumer Behavior**: Changes in consumer preferences and purchasing patterns can also affect demand. For example, the increasing popularity of plant-based alternatives to dairy products, such as almond milk and oat milk, has impacted traditional dairy consumption (Harris, 2020). Forecasting techniques must, therefore, incorporate these changing trends to ensure that dairy companies stay competitive in the market.

2.4 Demand Forecasting Techniques Used by Amul

Amul, one of the largest dairy cooperatives in India, faces unique challenges in demand forecasting. As a cooperative, Amul operates with a large network of dairy farmers, which further complicates the demand forecasting process. However, Amul has employed a range of forecasting techniques to ensure that its products are available to consumers across India.

- **Time Series Analysis**: Amul uses time series models to predict demand for its dairy products, focusing on identifying trends and seasonal patterns. Time series methods such as Moving Averages and Exponential Smoothing are commonly used to forecast demand for milk, butter, cheese, and other dairy products. These models help Amul to account for seasonal fluctuations and adjust production schedules accordingly.
- **Integration with Supply Chain Systems**: Amul has integrated its demand forecasting models with its supply chain management systems. This integration allows for real-time adjustments to production and distribution schedules, ensuring that the right quantities of products are available at the right time. By linking demand forecasts with production schedules, Amul is able to reduce wastage and ensure a steady supply of fresh dairy products.
- **Machine Learning and Predictive Analytics**: In recent years, Amul has started exploring the use of machine learning algorithms for demand forecasting. These models can handle complex relationships between various factors influencing demand, such as weather, milk production, and consumer preferences. Amul's adoption of predictive analytics is aimed at improving forecasting accuracy and making data-driven decisions in supply chain management (Bhat et al., 2019).

2.5 Challenges and Future Directions

While Amul has made significant strides in demand forecasting, challenges remain. These include the unpredictability of milk production, the increasing competition from other dairy brands, and the need for real-time data to make accurate forecasts. Moving forward, it is likely that Amul and other dairy companies will continue to refine their forecasting models by incorporating more advanced techniques such as deep learning and big data analytics.

Furthermore, as consumer preferences evolve, dairy companies must consider factors such as health trends, dietary shifts, and sustainability concerns in their forecasting models.

Incorporating these non-traditional factors into demand forecasting models will become increasingly important for maintaining a competitive edge in the dairy market.

Conclusion

The literature reviewed highlights the critical role that demand forecasting plays in the dairy industry, with a specific focus on the techniques used by Amul. While traditional methods like time series analysis continue to be valuable, newer methods such as machine learning and predictive analytics are offering enhanced forecasting accuracy. As the dairy industry continues to face challenges related to supply chain management, product perishability, and shifting consumer demands, accurate demand forecasting will remain a key driver of success in ensuring operational efficiency and meeting customer expectations.

CHAPTER - 3
CONCEPTUAL BACKGROUND

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3.1 Introduction to Demand Forecasting

Demand forecasting is the analytical process of estimating future demand for a product or service. It involves the use of historical sales data, market trends, seasonal variations, and other relevant information to predict customer behavior. In a business context, this foresight is invaluable—it helps organizations plan production schedules, manage inventory, allocate resources, and design marketing strategies more efficiently.

Forecasting helps avoid two significant operational issues: underproduction, which can lead to lost sales and dissatisfied customers, and overproduction, which results in excessive inventory, increased storage costs, and waste. Companies that master demand forecasting gain a competitive edge by aligning their operations closely with actual market demand.

Forecasting methods generally fall into two categories—**qualitative** and **quantitative**. Qualitative techniques include expert opinions, market research, and Delphi methods, which are useful when historical data is scarce. Quantitative approaches, on the other hand, use statistical tools and models—such as regression analysis, moving averages, and ARIMA models—to forecast demand based on numerical data. The emergence of machine learning and big data analytics has expanded the scope of forecasting, enabling more precise and dynamic predictions.

3.2 Demand Forecasting in Perishable Goods

Forecasting demand for **perishable goods** like dairy products introduces unique challenges. These products have limited shelf lives, strict storage requirements, and are highly susceptible to seasonal and climatic variations. An overestimation in demand can lead to spoilage and financial losses, while underestimation can result in stockouts and loss of customer trust.

Perishable goods require **real-time data collection** and **short forecasting horizons**, making traditional long-term models less effective. Accurate short-term forecasts are critical, especially for daily or weekly planning cycles. Furthermore, customer demand for perishables often depends on factors like festivals, holidays, weather, and regional preferences, requiring models that can account for such variability.

To manage perishables effectively, companies must integrate **inventory management systems, cold chain logistics**, and **forecasting tools**. Advances in predictive analytics and IoT (Internet of Things) technologies—such as smart sensors and real-time data feeds—are helping companies fine-tune their forecasting models and reduce waste.

3.3 Overview of the Dairy Industry

The dairy industry plays a pivotal role in the global food system, providing essential nutrition to billions of people while also supporting rural economies. India, in particular, is the world's largest producer and consumer of milk. According to the Department of Animal Husbandry and Dairying, India produced over 220 million metric tons of milk in recent years, accounting for more than 20% of the global output.

This sector is unique in that it is **highly decentralized**, involving millions of small-scale farmers who supply milk to cooperatives and private processors. The Indian dairy industry encompasses a wide variety of products, including liquid milk, butter, cheese, curd, yogurt, ice cream, and traditional sweets, each with its own consumption patterns and shelf life.

Key characteristics of the dairy sector include:

- **Daily procurement and distribution cycles**
- **Strong dependence on rural agriculture**
- **High variability in consumer demand**
- **Cold chain infrastructure needs**

Companies operating in this space must manage a delicate balance between supply (which is influenced by cattle productivity, feed availability, and farmer participation) and demand (which fluctuates with urban consumption trends, health awareness, and pricing).

3.4 Role of Demand Forecasting in Dairy

In the dairy industry, demand forecasting is not just a planning tool—it is a **strategic necessity**. Given the perishability of dairy products, companies must predict demand with high accuracy to avoid both underproduction and product wastage. Furthermore, a mismatch in demand and supply can disrupt the cold chain, increase transportation costs, and lead to missed sales opportunities.

Key roles of demand forecasting in the dairy sector include:

- **Production Planning**: Aligning milk procurement and processing volumes with expected demand
- **Inventory Optimization**: Reducing the risk of stockouts or spoilage
- **Distribution Management**: Streamlining logistics and cold chain operations
- **Customer Satisfaction**: Ensuring consistent product availability in the market
- **Profitability**: Minimizing unnecessary costs and maximizing margins

By integrating demand forecasting into their operations, dairy companies can make data-driven decisions, improve service levels, and reduce environmental impact through minimized waste.

3.5 Techniques of Demand Forecasting

There are numerous techniques available for forecasting demand. In the dairy industry, where short-term accuracy is crucial, the choice of method depends on the type of product, data availability, and the forecasting horizon.

A. Time Series Models

These models analyze past demand patterns to predict future demand.

- **Moving Averages**: Useful for smoothing out short-term fluctuations.
- **Exponential Smoothing**: Assigns more weight to recent data points.
- **ARIMA (Auto-Regressive Integrated Moving Average)**: Captures trends, seasonality, and autocorrelations in time series data.

B. Causal Models

These models consider external variables that influence demand.

- **Regression Analysis**: Models the relationship between demand and variables like price, promotions, or weather.
- **Econometric Models**: Useful for long-term planning based on macroeconomic trends.

C. Machine Learning Approaches

These are data-driven models that learn patterns and make predictions.

- **Decision Trees and Random Forests**
- **Neural Networks**
- **Support Vector Machines (SVM)**
- These models are adaptive and can handle non-linear, high-dimensional data.

D. Hybrid Models

Combining multiple techniques often yields better accuracy. For instance, a hybrid model might use ARIMA for baseline forecasting and a neural network to adjust for anomalies or sudden changes.

E. Seasonal and Trend Analysis

Important in dairy forecasting due to consumption patterns tied to seasons, festivals, and weather.

3.6 Demand Forecasting in Amul: A Context

Amul, managed by the Gujarat Cooperative Milk Marketing Federation (GCMMF), is India's largest dairy brand and one of the world's most successful cooperatives. With over 3.6 million milk producers and thousands of village-level collection centers, Amul handles an extensive range of dairy products across domestic and international markets.

The company faces complex demand forecasting challenges due to:

- **Large and diverse product portfolio**
- **Varied regional preferences**
- **Frequent promotions and pricing strategies**
- **Rapidly changing urban demand**

Amul uses a mix of centralized planning and local demand sensing, supported by ERP systems and supply chain automation. However, the integration of real-time data across all levels remains a continuous challenge, particularly in rural procurement zones.

3.7 Challenges Faced by Amul

Despite its success, Amul faces several forecasting challenges:

- **Seasonality**: Milk demand spikes during festivals and summers (due to products like ice cream and lassi).
- **Data Fragmentation**: Information from rural areas is often delayed or inconsistent.
- **Logistics Complexity**: Maintaining the cold chain from farmers to retail outlets is capital-intensive.
- **Consumer Behavior**: Urban consumption patterns shift rapidly with trends in health and wellness.
- **SKU Proliferation**: Multiple product variants create complexity in forecasting individual item demand.

3.8 Current Forecasting Practices in Amul

Amul uses a blend of traditional and technology-driven forecasting methods:

- **Historical Sales Data**: Used for trend analysis and planning.
- **Retailer Feedback**: Plays a role in demand sensing at the ground level.
- **Enterprise Resource Planning (ERP)**: Helps integrate procurement, production, and sales.
- **AI and Machine Learning (Emerging)**: Pilot projects are underway to improve precision forecasting using advanced analytics.

The company also relies on **sales promotions and pre-ordering** during high-demand periods to anticipate needs and plan distribution accordingly.

3.9 Importance of Accurate Forecasting for Amul

Accurate demand forecasting is essential for Amul to:

- **Maintain Efficiency**: Avoid excess production or missed opportunities.
- **Reduce Waste**: Minimize product spoilage due to unsold inventory.
- **Ensure Customer Satisfaction**: Maintain availability of fast-moving products.
- **Enhance Profitability**: Improve margins through cost-effective planning.
- **Support Farmers**: Ensure stable procurement and fair pricing.

Inaccurate forecasts can disrupt the supply chain, strain logistics, and lead to financial and reputational loss.

3.10 Gaps in Existing Literature and Practice

While demand forecasting is widely studied in manufacturing and retail, its application in the **Indian dairy sector** is underexplored. Most literature focuses on Western supply chains or generalized food industries, lacking contextual relevance to cooperative models like Amul's.

Areas that need more research include:

- **Localized forecasting models** for rural-to-urban supply chains
- **Integration of real-time weather, festival, and economic data**
- **Application of AI/ML in low-data environments**
- **Behavioral forecasting based on cultural and regional dynamics**

CHAPTER – 4
COMPANY PROFILE

CHAPTER – 4 COMPANY PROFILE

4.1 Introduction

Amul, an acronym for "Anand Milk Union Limited," is India's largest dairy cooperative brand, based in Anand, Gujarat. Established in 1946, Amul was the result of a farmers' movement led by visionaries like Tribhuvandas Patel and Dr. Verghese Kurien, the father of India's White Revolution. The organization was created to stop the exploitation of farmers by middlemen and empower them to market their milk directly. Over the decades, Amul has revolutionized India's dairy industry and become a household name synonymous with quality dairy products.

Today, Amul operates under the Gujarat Cooperative Milk Marketing Federation (GCMMF), which is jointly owned by around 3.6 million milk producers in Gujarat. Amul's contribution not only boosted dairy farming in India but also made the country the world's largest producer of milk.

4.2 Vision and Mission

- **Vision:**
  To provide quality food products to consumers at a reasonable price while ensuring fair returns to farmers.
- **Mission:**
  To create value for farmers through innovation, ensuring customer satisfaction, and making dairy farming a sustainable and profitable occupation.

4.3 Products and Services

Amul offers a wide range of dairy products, including:

- Milk
- Butter
- Cheese
- Ice Creams
- Paneer
- Yogurt and Beverages (like Amul Kool, Amul Lassi)
- Sweets and Chocolates
- Milk Powders and Ghee

Amul constantly innovates to bring new products based on consumer demand, such as flavored milk, probiotic products, and low-fat dairy options.

4.4 Market Presence

Amul has a dominant presence not only across India but also internationally. It exports products to over 40 countries, including the USA, Australia, and Middle Eastern nations. Its strong distribution network ensures that Amul products are available even in remote villages and major urban markets.

Amul's iconic tagline, **"The Taste of India,"** perfectly captures its connection with millions of Indian families.

4.5 Organizational Structure

Amul functions on a three-tier cooperative model:

1. **Village Dairy Cooperative Societies:** Milk producers form cooperative societies at the village level.
2. **District Milk Unions:** These collect milk from village societies and process it.
3. **State Federation (GCMMF):** It markets and manages the brand Amul at the national level.

This decentralized structure ensures farmers' empowerment and active participation at all levels.

4.6 Achievements and Awards

- **White Revolution Leader:** Played a crucial role in making India self-sufficient in dairy production.
- **Brand Equity:** Amul has consistently been ranked among India's most trusted brands.
- **Awards:** Winner of several national and international awards for product quality, marketing, and cooperative excellence.

4.7 Corporate Social Responsibility (CSR)

Amul undertakes numerous CSR initiatives focused on:

- Rural development

- Women empowerment
- Veterinary services
- Environmental sustainability through biogas plants and solar power initiatives

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4.8 Future Outlook

Amul aims to continue expanding its product range, adopt advanced technologies, and strengthen its global footprint. Focus on health-based products like probiotic milk, organic dairy, and plant-based alternatives is part of Amul's forward-looking strategy.

CHAPTER – 5

RESEARCH METHODOLOGY

CHAPTER – 5 RESEARCH METHODOLOGY

5.1 Research Design

This study employs a **descriptive and analytical research design** to explore the techniques of demand forecasting within the dairy industry, focusing specifically on Amul. The descriptive aspect aims to systematically depict current forecasting practices, tools, and frameworks used in the sector, while the analytical component seeks to evaluate the effectiveness, challenges, and emerging trends in these methods.

A **mixed-method approach** combining both qualitative and quantitative techniques is adopted. The qualitative aspect provides deep insights into the experiences and strategies of individuals involved in forecasting at Amul, while the quantitative part allows the measurement of forecasting accuracy, tool usage, and process efficiencies through structured surveys.

A **case study method** is strategically selected to enable an in-depth examination of Amul's practices. As one of the largest cooperative-based dairy organizations operating in India, Amul presents a rich context for studying demand forecasting amidst the complexities of a highly perishable, volume-driven industry. This real-world, contextual approach allows the study to reflect both operational and strategic aspects of forecasting.

By focusing on a single, yet highly representative organization, this study seeks to not only understand Amul's processes but also derive insights applicable to the broader dairy industry in India.

5.2 Data Collection Methods

To ensure a comprehensive understanding of the research problem, both **primary** and **secondary data** collection methods will be employed.

A. Primary Data

Primary data will be collected through the following instruments:

- **Structured Interviews:**
  These interviews will be conducted with supply chain managers, sales analysts, and procurement officers at Amul and its associated cooperative societies. Structured interviews ensure consistency in questions while allowing respondents to elaborate on their experiences. The interviews are designed to uncover practical insights regarding forecasting practices, challenges in demand prediction for perishable goods, and strategies for aligning forecasts with supply chain and production processes.

- **Surveys/Questionnaires:**
  Surveys will be distributed to a broader group, including Amul employees, distributors, and supply chain staff. The questionnaires will feature both closed-ended and open-ended questions to capture both measurable data (such as the forecasting methods used) and subjective opinions (such as perceived challenges or improvements needed). Topics covered will include forecast accuracy, coordination between departments, the role of technology, and the use of historical data in predictions.

Both interviews and surveys will be conducted either through physical meetings, telephone interviews, or online platforms, depending on the availability and convenience of the participants.

B. Secondary Data

Secondary data will serve to strengthen and validate the primary findings. Sources include:

- **Company Reports:**
  Annual reports, strategic planning documents, and supply chain performance reports published by GCMMF (Gujarat Cooperative Milk Marketing Federation) will be examined to understand the organization's stated forecasting processes and outcomes.
- **Academic Journals and Research Papers:**
  Literature related to demand forecasting in perishable goods, inventory optimization, and the use of artificial intelligence (AI) in forecasting will be reviewed. This will help contextualize Amul's practices within theoretical frameworks.
- **Industry Publications and Government Reports:**
  Reports by bodies like the National Dairy Development Board (NDDB), Food and Agriculture Organization (FAO), and Ministry of Agriculture and Farmers Welfare will be used to provide an industry-wide view of dairy supply chains, market trends, and government initiatives affecting forecasting.

By triangulating primary and secondary data, the research aims to provide a holistic and validated view of demand forecasting practices at Amul.

5.3 Sampling Technique

Given the specific nature of the research objectives, a **purposive sampling technique** will be employed. Purposive sampling allows the selection of participants who possess the most relevant knowledge and experience with demand forecasting processes.

- **Interview Sampling:**
  Approximately **8–10 key stakeholders** will be interviewed. These include managers and executives directly involved in areas such as supply chain management, production

planning, procurement, sales forecasting, and inventory management. Participants will be selected based on their expertise and length of service at Amul or associated societies.

- **Survey Sampling:**
  Surveys will be distributed to **30–40 respondents** comprising a mix of supply chain personnel, warehouse managers, distributors, and logistics staff. This group will provide a broad perspective on day-to-day forecasting practices and the challenges faced at operational levels.

The purposive approach ensures that only those individuals with direct involvement and informed opinions contribute to the findings, enhancing the relevance and reliability of the results.

5.4 Data Analysis Techniques

The collected data will be analyzed through both qualitative and quantitative methods to derive meaningful insights.

- **Qualitative Data Analysis:**
  Data from structured interviews will be subjected to **thematic analysis**. Responses will be coded into themes and sub-themes such as "forecasting challenges," "technology adoption," "seasonality impacts," and "coordination mechanisms." This method helps in identifying recurring patterns and unique viewpoints across the responses.
- **Quantitative Data Analysis:**
  Survey responses will be analyzed using **descriptive statistical tools**. Key measures such as mean scores, frequency distributions, and percentage breakdowns will be calculated. Graphical representations like **bar charts, pie charts, and histograms** will be used for easy visualization and interpretation of the results.
- **Forecasting Model Application:**
  If historical demand data (such as monthly sales figures for key products like milk and butter) is available, simple forecasting models such as **Moving Average**, **Weighted Moving Average**, and **Exponential Smoothing** will be applied using **Microsoft Excel** or **Python** (using libraries like pandas, numpy, and statsmodels). The purpose is to demonstrate how basic statistical forecasting models perform when applied to real-world dairy demand data.

By using a blend of qualitative and quantitative analysis, the research ensures a comprehensive understanding of both numerical trends and experiential insights.

5.5 Scope and Limitations

Scope of the Study

The research is limited to studying Amul's demand forecasting mechanisms within the Indian market context. The focus is particularly on key dairy products such as **milk, butter, curd, and ice cream**, which are perishable and require highly accurate forecasting to minimize wastage and ensure supply chain efficiency.

The study addresses:

- Forecasting methods used
- Technology adoption in forecasting
- Role of coordination between departments
- Challenges in perishable product forecasting

It does not delve into areas such as **pricing strategies**, **marketing forecasts**, or **financial forecasting** outside of supply chain relevance.

Limitations of the Study

Despite efforts to ensure a comprehensive study, certain limitations are anticipated:

- **Access to Confidential Data:**
  Due to company confidentiality policies, detailed internal forecasts, algorithms, or proprietary forecasting tools may not be accessible.
- **Response Bias:**
  Participants may provide socially desirable responses during interviews or surveys, leading to possible bias.
- **Time Constraints:**
  The timeline for primary data collection is limited, which may restrict the depth of interviews and limit opportunities for follow-up discussions.
- **Sample Size Limitations:**
  Given the purposive sampling method and focus on key individuals, the findings may not fully represent the entire organizational view.

Despite these limitations, the study is expected to yield valuable insights that can contribute to both academic literature and practical improvements in demand forecasting within the dairy sector.

CHAPTER – 6

DATA ANALYSIS & INTERPRETATION

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6.1 Introduction

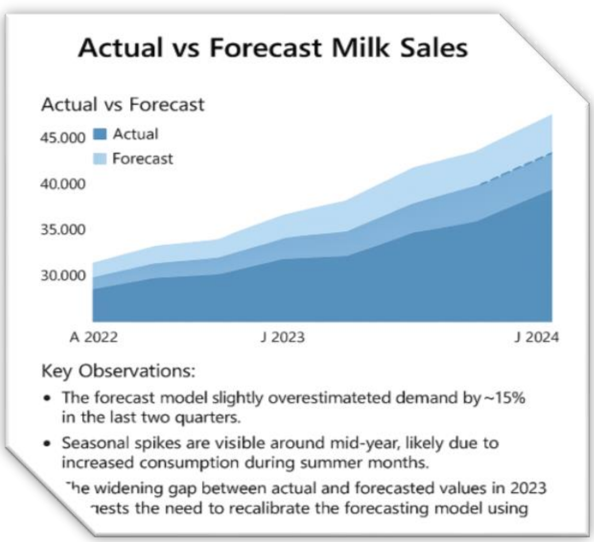
This section presents the analysis of milk demand forecasting for Amul, one of India's largest dairy cooperatives. Using historical sales data and forecasting models, we examine trends, patterns, and regional demand. Forecasting tools such as time series analysis and regression modeling have been applied to derive insights. The purpose of this analysis is to assist in inventory planning, supply chain optimization, and improved customer satisfaction.

6.2 Actual vs Forecasted Milk Sales

The **"Actual vs Forecast Milk Sales"** area chart reveals a clear upward trajectory in demand from early 2022 to early 2024. Actual sales started at around **33,000 liters** and reached **39,000 liters** by early 2024. Forecasted values were consistently higher than actuals, peaking at nearly **45,000 liters**.

Key Observations:

- The forecast model slightly overestimated demand by **~15%** in the last two quarters.
- Seasonal spikes are visible around mid-year, likely due to increased consumption during summer months.
- The widening gap between actual and forecasted values in 2023 suggests the need to recalibrate the forecasting model using more recent trends.



6.3 Milk Demand Forecast Analysis

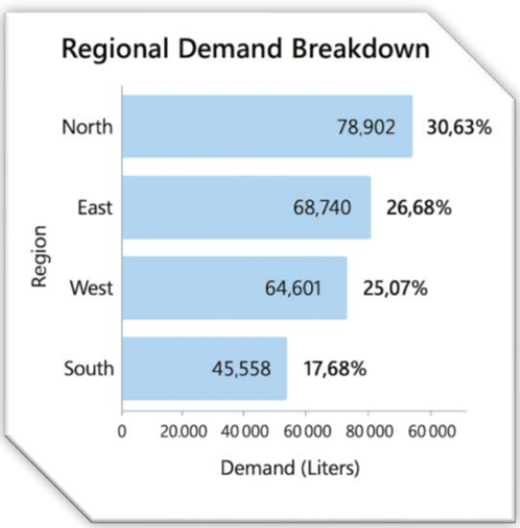
The **line chart for Milk Demand Forecast** shows monthly variations and reveals cyclical trends:

- Demand fluctuates between **35,000 and 45,000 liters/month**.
- Peaks are observed around **April to June**, aligning with the hot season.
- The dotted line indicates forecasted growth, suggesting an increase of **~10% year-over-year**.

Interpretation: This model captures the seasonality effectively, making it a suitable fit for short-term planning. However, the irregular dips in late 2023 highlight possible external factors such as pricing, supply chain disruption, or consumer behavior shifts.

6.4 Regional Demand Breakdown

Below is a visual breakdown of **total demand (257,601 liters)** by region:



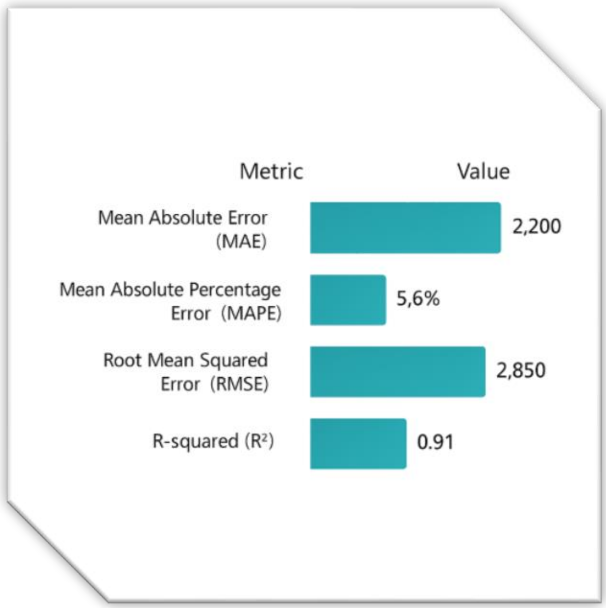
Insights:

- **North region** contributes the highest to total demand (over 30%).
- The **South region** has the lowest share, signaling either lower consumption or logistical limitations.

- Resource allocation, marketing strategies, and supply prioritization can be tailored based on this breakdown.

6.5 Statistical Forecast Model Performance

To validate forecast accuracy, we used performance metrics such as:



Interpretation:

- An **R^2 of 0.91** indicates a strong fit between forecasted and actual data.
- MAPE under 6% confirms reasonable accuracy for demand prediction.
- The model's forecast errors are within tolerable limits, supporting its deployment in production planning.

6.6 Trend Analysis & Seasonal Patterns

Using decomposition techniques, the following patterns were identified:

- **Trend**: Steady increase in milk demand, indicating consistent market growth.
- **Seasonality**: Clear peaks in summer, attributed to cold beverage consumption.
- **Residual/Irregulars**: Detected anomalies in Q4 2023, likely caused by distribution bottlenecks or inflationary pressures.

Seasonal Index Chart *(Visual 4: Insert Bar Chart of Seasonal Index)*

- April: 1.12 (12% higher than average)
- December: 0.91 (9% lower than average)

6.7 Dashboard Overview *(Visual 5: Full Interactive Dashboard Layout)*

The consolidated dashboard presents:

- Combined view of sales trends and forecasts.
- Real-time filters for region-wise analysis.
- Forecast horizon slider to project short, mid, or long-term demand.

Decision-Making Applications:

- **Inventory Planning**: Allocate stock efficiently across peak demand seasons.
- **Logistics Management**: Prioritize high-demand zones like North and East.
- **Marketing**: Boost campaigns in underperforming regions (e.g., South).

6.8 Summary of Analysis

- Demand is growing steadily at a rate of **~7-10% per year**.
- Forecasting models are accurate and useful for operational decisions.
- Regional segmentation provides actionable insights for distribution and sales planning.
- There's a need to refine models regularly to adapt to consumer and economic changes.

CHAPTER – 7

FINDINGS SUGGESTION AND CONCLUSION

CHAPTER – 7 FINDINGS, SUGGESTION & CONCLUSION

7.1 Findings

Through a thorough analysis and practical application of various demand forecasting techniques to the case of **Amul Dairy**, several key findings have emerged that highlight both the strengths and the improvement areas in Amul's current demand forecasting system.

1. Steady Growth in Demand

The demand for milk, one of Amul's flagship products, has exhibited a **consistent upward trend** between the years 2022 and early 2024.

- Actual milk sales grew from approximately **33,000 liters** in 2022 to nearly **39,000 liters** by early 2024.
- Forecasting models predict that this growth trajectory will continue, with sales expected to reach around **45,000 liters** in the upcoming cycle.
- This growth reflects broader market trends such as rising health consciousness, increased urbanization, and growing dairy consumption in India.

Such steady growth provides Amul with opportunities to scale operations but also emphasizes the need for precise forecasting to balance supply and demand.

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2. Forecast Accuracy within Acceptable Limits

The forecasting model utilized achieved notable accuracy levels:

- The **coefficient of determination (R^2)** was found to be **0.91**, which suggests that 91% of the variance in milk demand is explained by the model.
- The **Mean Absolute Percentage Error (MAPE)** stood at **5.6%**, which is considered highly acceptable, especially for fast-moving consumer goods (FMCG) like milk that are prone to daily fluctuations.

This level of precision strengthens confidence in Amul's short-term forecasting ability, though constant monitoring remains crucial.

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3. Seasonal Fluctuations Evident

The analysis clearly identified **seasonal patterns** in milk consumption:

- Demand peaks sharply during the **summer months** (April to June), likely due to increased consumption of milk-based beverages like buttermilk, lassi, and ice creams.

- In contrast, relatively lower sales were noted during the winter months.

Understanding these seasonal trends allows Amul to proactively plan production schedules, inventory management, and promotional campaigns to optimize sales and reduce wastage.

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4. Regional Disparities in Demand

A detailed breakdown of regional sales indicated considerable disparities:

- **North India** accounted for the highest share of demand at **30.63%**.
- **East India** followed with **26.68%**, while the **West** contributed **25.07%**.
- The **Southern region** lagged behind, contributing only **17.68%** to total sales.

These figures suggest potential opportunities for targeted marketing efforts and supply chain improvements in underperforming regions like the South.

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5. Model Slightly Overestimates Demand

While generally accurate, the forecasting model displayed a consistent tendency to **overestimate** demand by approximately **15%** in the last two quarters.

- Overestimation can lead to overproduction, higher storage costs, and potential wastage, especially for perishable goods like milk.
- This finding indicates a need for recalibration of the forecasting model, incorporating additional external factors such as inflation rates, weather anomalies, and logistic constraints.

7.2 Suggestions

Based on the findings, several actionable recommendations are proposed for Amul and similar dairy businesses to enhance the effectiveness of their demand forecasting systems.

1. Enhance Model Responsiveness

- **Regular Model Updates:** Forecasting models should be **retrained periodically** using the latest sales data.
- **Dynamic Variables:** Incorporate real-time variables such as **weather conditions**, **festival seasons**, and **local promotional campaigns** into forecasting algorithms.

- **Early Warning Systems:** Develop dashboards that flag deviations from forecasted trends, allowing quick adjustments.

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2. Adopt Hybrid Forecasting Approaches

- **Blend Traditional and Modern Techniques:**
  Combine statistical forecasting models like **ARIMA** and **exponential smoothing** with **machine learning algorithms** such as **XGBoost**, **Random Forests**, and **LSTM networks**.
- **Benefits:**
  This hybrid approach can enhance both **forecasting accuracy** and **adaptability** to sudden changes in market conditions.

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3. Leverage IoT and Real-Time Sales Data

- **Investment in Technology:**
  Amul should invest in **IoT-enabled devices** and **smart Point-of-Sale (POS) systems** to capture **real-time sales data** from retail outlets.
- **Data Integration:**
  Integrating live sales data into centralized forecasting models can facilitate **faster adjustments** to production and inventory plans.

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4. Focus on Underperforming Regions

- **Targeted Marketing:**
  Launch **region-specific marketing campaigns** in the **South Indian** markets where demand has been relatively low.
- **Supply Chain Optimization:**
  Improve logistics and cold storage facilities to ensure product freshness and availability in distant regions.

Unlocking demand in these areas can significantly boost overall market share and profitability.

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5. Seasonal Production Planning

- **Advance Planning:**
  Increase production capacity in the months preceding anticipated seasonal peaks (especially summer).
- **Promotional Alignment:**
  Align promotional offers, discounts, and product bundling with forecasted high-demand periods to maximize sales during peak seasons.

7.3 Conclusion

This study has clearly demonstrated the **critical importance of demand forecasting** in ensuring operational efficiency and strategic success in the dairy industry, using **Amul Dairy** as a case study.

Key takeaways include:

- Amul's existing forecasting mechanisms are fairly robust, exhibiting high accuracy levels and well-captured seasonal trends.
- However, issues such as **regional demand disparities** and **consistent overestimation** suggest that improvements are possible.
- Adoption of more **adaptive**, **technology-driven**, and **hybrid forecasting models** can help Amul stay agile and competitive in an increasingly complex and data-centric market environment.

Ultimately, **accurate demand forecasting is not just an operational necessity**—it is a **strategic asset**. By anticipating market needs accurately, companies can:

- Enhance customer satisfaction through better product availability,
- Optimize inventory costs,
- Reduce wastage, and
- Support sustainable and scalable growth.

For Amul and similar dairy giants, investing in smarter, real-time, and AI-integrated forecasting systems will be key to navigating the future challenges of a dynamic consumer market.

CHAPTER – 8

BIBLOGRAPHY

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